

Moaz Elborollosy

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Profile

Third-year Mechatronics Engineering student at German International University in Berlin (GPA 1.16, German scale) seeking Werkstudent roles in mechanical design, CAD, robotics, automation, or hardware development. Technical skills include SolidWorks, AutoCAD, 3D printing, Arduino, circuit design, and C/C++.

Education

German International University

Berlin, Germany

B.Sc. in Mechatronics Engineering | GPA: 1.16 (German scale, Grade A)

Sep. 2024 – Jul. 2028 (Expected)

- **Full Merit Scholarship:** Selected based on top academic standing to complete undergraduate studies in Germany.
- **Relevant Coursework:** Kinematics & Robotics, Control Systems, Signals & Systems, Embedded Systems, Microcontroller Programming, Computer Architecture.

Technical Skills

CAD & Prototyping: SolidWorks, AutoCAD, 3D Printing & Prototyping

Robotics & Automation: ROS 2, Arduino, PIC, Circuit Design, Signal Processing

Programming: C/C++, Python, Java, Assembly, HTML

Developer Tools: Linux (Ubuntu), Git, GitHub, VS Code

Languages: English (C1), German (A2), Arabic (Native)

Projects

Custom Quadcopter Drone | SolidWorks

[LinkedIn](#)

- Designed and assembled a complete multirotor drone in SolidWorks, integrating the airframe, electronics, and flight-controller packaging within tight spatial constraints.
- Used SolidWorks motion studies to verify propeller rotation, component clearances, and mechanical tolerances across the assembly.

Mechanical Robotic Gripper Assembly | SolidWorks

[LinkedIn](#)

- Designed a linkage-driven robotic gripper in SolidWorks and configured kinematic mates to model coordinated grasping motion and validate the mechanism's movement.

Embedded Robotic Arm | SolidWorks, C++, Arduino

- Designed a multi-axis robotic arm in SolidWorks, defining the mechanical structure and joint configuration from concept through full CAD assembly.
- Programmed Arduino C++ control logic to perform kinematic calculations and coordinate synchronized servo motion across multiple joints.

Low-Pass Filter Circuit | Circuit Design & Simulation

- Designed and simulated an active low-pass filter circuit, verifying high-frequency noise attenuation while preserving signal integrity.

Assembly Language Interpreter | Java

[GitHub](#)

- Built a Java assembly-language interpreter from scratch to parse, decode, and execute low-level instructions while emulating CPU cycles and memory management.
- Passed all required and bonus test cases; our team achieved the highest overall project score among all teams.

Object-Oriented Desktop Game | Java

[GitHub](#)

- Developed the game engine and core gameplay logic in Java using object-oriented programming, including collision detection and exception handling for reliable execution.
- Designed and implemented the complete GUI, delivering smooth user interaction and stable, responsive gameplay in the final game.